

REVIEW OF LITERATURE

Anatomical Variation of Paranasal Air Sinuses: A CT Study

1. INTRODUCTION TO PARANASAL SINUSES

The paranasal sinuses (PNS) are air-filled cavities within the bones of the skull and face that communicate with the nasal cavity. These include the maxillary, frontal, ethmoid, and sphenoid sinuses. Understanding their anatomical variations is crucial for otorhinolaryngologists, radiologists, and anatomists, particularly in the context of functional endoscopic sinus surgery (FESS) and other surgical interventions.

1.1 Historical Background

The study of paranasal sinuses dates back to ancient times. Galen (130-200 AD) first described the frontal and maxillary sinuses. Leonardo da Vinci (1489) illustrated the maxillary sinus in his anatomical drawings. Nathaniel Highmore (1651) provided detailed descriptions of the maxillary sinus, which is eponymously named the "Antrum of Highmore."

Messerklinger (1978) revolutionized the understanding of sinus physiology by introducing the concept of mucociliary clearance and the osteomeatal complex (OMC), laying the foundation for functional endoscopic sinus surgery.

2. EMBRYOLOGY OF PARANASAL SINUSES

Sadler (2012) described that paranasal sinuses develop as outgrowths from the nasal cavity into surrounding bones during fetal and postnatal life:

- Maxillary sinus: Begins developing at 3rd month of fetal life; pneumatization continues until 12-18 years
- Ethmoid sinus: Present at birth; completes development by 12-14 years
- Frontal sinus: Appears at 2 years of age; fully develops by 20 years
- Sphenoid sinus: Pneumatization begins at 3 years; completes by 12-14 years

Stammberger (1991) emphasized that variations in development can lead to significant anatomical differences affecting surgical approaches.

3. ANATOMY OF PARANASAL SINUSES

3.1 Maxillary Sinus

The maxillary sinus is the largest paranasal sinus, located within the body of maxilla. According to Standring (2016, Gray's Anatomy), it has:

- Average volume: 15 ml in adults
- Shape: Pyramidal with apex towards zygomatic process
- Relations: Orbit (superior), nasal cavity (medial), alveolar process (inferior), infratemporal fossa (posterior)

3.2 Ethmoid Sinus

Stammberger (1991) described ethmoid sinuses as 3-18 air cells divided into:

- Anterior ethmoid cells (drain into middle meatus)
- Posterior ethmoid cells (drain into superior meatus)

3.3 Frontal Sinus

Located within the frontal bone, Lang (1989) documented:

- Highly variable in size and shape
- May be absent (aplasia) in 5-15% of population
- Asymmetry is common

3.4 Sphenoid Sinus

Hamberger et al. (1961) classified sphenoid sinus pneumatization into:

- Conchal type (2-3%)
- Presellar type (11-24%)
- Sellar type (73-86%)

4. OSTEOMEATAL COMPLEX (OMC)

The osteomeatal complex is a functional unit critical for sinus drainage. Stammberger and Kennedy (1995) defined it as including:

- Maxillary sinus ostium
- Ethmoid infundibulum
- Hiatus semilunaris
- Middle turbinate
- Uncinate process
- Ethmoid bulla

Zinreich et al. (1988) demonstrated through CT studies that OMC obstruction leads to sinus disease.

5. ANATOMICAL VARIATIONS OF PARANASAL SINUSES

5.1 Nasal Septum Variations

5.1.1 Deviated Nasal Septum (DNS)

Earwaker (1993) reported DNS prevalence of 20-79% in CT studies. Stallman et al. (2004) found DNS in 65% of patients undergoing CT scans.

Indian Studies:

- Dua et al. (2005) reported DNS in 75.4% of Indian population
- Adeel et al. (2013) found 64.5% prevalence in their study
- Mamatha et al. (2015) documented 58% prevalence in South Indian population

5.1.2 Nasal Septal Spur

Bolger et al. (1991) reported septal spur prevalence of 30-50%.

5.2 Turbinate Variations

5.2.1 Concha Bullosa

Concha bullosa is pneumatization of the middle turbinate. Bolger et al. (1991) classified it into:

- Lamellar type (vertical lamella pneumatization)
- Bulbous type (inferior bulbous segment pneumatization)
- Extensive type (both lamellar and bulbous)

Prevalence Studies:

- Tonai and Baba (1996): 34% in Japanese population
- Stallman et al. (2004): 35.8%
- Adeel et al. (2013): 30.4%
- Sharma et al. (2016) (India): 32.5%

5.2.2 Paradoxical Middle Turbinate

Paradoxical curvature where convexity faces laterally instead of medially.

- Bolger et al. (1991): 26.1%
- Stallman et al. (2004): 26.9%
- Badia et al. (2005): 17%
- Adeel et al. (2013): 15%

5.2.3 Secondary Middle Turbinate

Khanobthamchai et al. (1991) first described accessory/secondary middle turbinate with prevalence of 0.5-2%.

5.3 Uncinate Process Variations

Stammberger (1991) described superior attachment variations of uncinate process:

1. Type I: Attaches to lamina papyracea
2. Type II: Attaches to skull base (fovea ethmoidalis)
3. Type III: Attaches to middle turbinate
4. Type IV: Combination attachments
5. Type V: Attaches to roof of maxillary sinus
6. Type VI: Free edge or bifurcated

Landsberg and Friedman (2001) found:

- Type I: 52%
- Type II: 6%
- Type III: 26%
- Combined types: 16%

5.3.1 Pneumatized Uncinate Process (Uncinate Bulla)

Kennedy and Zinreich (1988) reported 0.4-2.5% prevalence.

5.4 Ethmoid Sinus Variations

5.4.1 Agger Nasi Cells

Agger nasi cells are the most anterior ethmoid cells, located anterior and superior to the attachment of middle turbinate.

Prevalence:

- Bolger et al. (1991): 98.5%
- Stammberger (1991): 89%
- Wormald (2003): 96%
- Jain et al. (2012) (India): 92%

5.4.2 Haller Cells (Infraorbital Ethmoid Cells)

Ethmoid air cells projecting along the medial orbital floor, adjacent to maxillary sinus ostium.

Prevalence Studies:

- Bolger et al. (1991): 10.3%
- Stackpole and Edelstein (1997): 18%
- Kantarci et al. (2004): 16%
- Jain et al. (2012) (India): 15.8%
- Sharma et al. (2016) (India): 18.2%

5.4.3 Onodi Cells (Sphenoethmoid Cells)

Posterior ethmoid cells that extend superolateral to sphenoid sinus with close relation to optic nerve.

Prevalence:

- Weinberger (1996): 8-14%
- Delano et al. (1996): 12%
- Wormald (2003): 14%
- Jain et al. (2012) (India): 10.5%

Clinical Significance: Risk of optic nerve injury during FESS.

5.4.4 Frontal Cells (Kuhn Classification)

Bent and Kuhn (1994) classified frontal recess cells:

- Type I: Single cell above agger nasi
- Type II: Tier of cells above agger nasi
- Type III: Single cell extending into frontal sinus
- Type IV: Isolated cell within frontal sinus

Prevalence (Type III/IV):

- Meyer et al. (2003): 8-12%
- Wormald (2003): 6-8%

5.4.5 Supraorbital Ethmoid Cells

Ethmoid cells extending over the orbit.

- Bolger et al. (1991): 7.2%
- DelGaudio et al. (2005): 15%

5.5 Frontal Sinus Variations

5.5.1 Frontal Sinus Aplasia/Hypoplasia

- Aydinlioglu et al. (2003): Aplasia 5-15%
- Ponde et al. (2003) (India): Aplasia 4.2%
- Nikam et al. (2015) (India): Aplasia 3.8%

5.5.2 Frontal Sinus Asymmetry

- Harris et al. (1987): Asymmetry in 85% cases
- Guerram et al. (2014): Used for forensic identification

5.5.3 Interfrontal Septal Cell (Frontal Septal Cell)

Cell within the interfrontal septum.

- McLaughlin et al. (2001): 12-36%

5.6 Maxillary Sinus Variations

5.6.1 Maxillary Sinus Hypoplasia

- Sirikci et al. (2000): 1.7-10.4%
- Jain et al. (2012) (India): 4.3%

5.6.2 Maxillary Sinus Septa (Underwood's Septa)

Bony septa dividing maxillary sinus, first described by Underwood (1910).

- Velasquez-Plata et al. (2002): 21.6%
- Kim et al. (2006): 26.5%
- Maestre-Ferrin et al. (2011): 32.7%

5.6.3 Accessory Maxillary Ostium

Additional ostium in lateral nasal wall besides natural ostium.

- Van Alyea (1936): 23%
- Stammberger (1991): 10-30%
- Jain et al. (2012) (India): 18.4%

5.7 Sphenoid Sinus Variations

5.7.1 Sphenoid Sinus Pneumatization Types

Hammer and Radberg (1961) and Hamberger et al. (1961) classification:

- Conchal type: Minimal pneumatization (2-3%)
- Presellar type: Pneumatization up to anterior wall of sella (11-24%)
- Sellar type: Extends under sella turcica (73-86%)

Indian Studies:

- Vaishya et al. (2015): Sellar type 76%, Presellar 20%, Conchal 4%
- Naveen et al. (2018): Sellar type 82.5%

5.7.2 Sphenoid Sinus Septation

- Sethi et al. (1995): Single septum 68%, Multiple septa 32%
- Sareen et al. (2005) (India): Asymmetric septation 78%
- Hamid et al. (2008): Septum attachment to carotid canal 8-25%

5.7.3 Lateral Recess (Pterygoid Recess)

Extension of sphenoid sinus into pterygoid plates.

- Bolger et al. (1991): 39%
- Wang et al. (2010): 45%

5.7.4 Internal Carotid Artery Dehiscence

Absence of bony covering over carotid artery in sphenoid sinus.

- Fujii et al. (1979): 4-8%
- Rhoton (2002): 6-22%
- Vaishya et al. (2015) (India): 7.5%

Clinical Significance: Risk of catastrophic hemorrhage during transsphenoidal surgery.

5.7.5 Optic Nerve Dehiscence/Protrusion

- DeLano et al. (1996): Protrusion 40%, Dehiscence 6%
- Sirikci et al. (2000): Protrusion 8-40%

5.8 Fovea Ethmoidalis (Roof of Ethmoid) Variations

Keros (1962) classified depth of olfactory fossa:

- Type I: 1-3 mm depth (11-26%)
- Type II: 4-7 mm depth (52-73%)
- Type III: 8-16 mm depth (8-26%)

Indian Studies:

- Jain et al. (2012): Type I 15%, Type II 70%, Type III 15%
- Sharma et al. (2016): Type I 12%, Type II 68%, Type III 20%

Clinical Significance: Higher Keros type indicates increased risk of skull base injury during FESS.

5.9 Cribriform Plate Asymmetry

- Meloni et al. (1992): Asymmetry in 10-20%
- Lebowitz et al. (2001): Asymmetry increases surgical risk

6. CT IMAGING IN PARANASAL SINUS EVALUATION

6.1 Role of CT in Sinus Anatomy

Zinreich et al. (1987) established CT as the gold standard for evaluating paranasal sinus anatomy and pathology.

Advantages of CT:

- Superior bone detail
- Multiplanar reconstruction capability
- Assessment of anatomical variations
- Pre-surgical planning

6.2 CT Protocol for Paranasal Sinuses

Lund-Mackay (1993) standardized CT evaluation and scoring system for chronic rhinosinusitis:

- Each sinus scored 0-2 based on opacification
- OMC scored 0 or 2
- Maximum score: 24 (bilateral)

6.3 Multidetector CT (MDCT) Advances

Bisdas et al. (2004) demonstrated advantages of MDCT:

- Thinner slices (0.5-1 mm)
- Better multiplanar reformations
- Reduced artifacts
- Lower radiation dose protocols

7. INDIAN STUDIES ON PARANASAL SINUS VARIATIONS

7.1 Regional Variations in India

Dua et al. (2005) - North India:

- DNS: 75.4%

- Concha bullosa: 31%
- Agger nasi cells: 87%

Jain et al. (2012) - Central India:

- DNS: 68.4%
- Concha bullosa: 28.9%
- Agger nasi cells: 92%
- Haller cells: 15.8%
- Onodi cells: 10.5%

Sharma et al. (2014) - Western India:

- DNS: 62%
- Concha bullosa: 32.5%
- Haller cells: 18.2%

Mamatha et al. (2015) - South India:

- DNS: 58%
- Concha bullosa: 25%
- Paradoxical middle turbinate: 12%

Adeel et al. (2013) - Multi-center India:

- DNS: 64.5%
- Concha bullosa: 30.4%
- Paradoxical turbinate: 15%
- Agger nasi: 89%

7.2 Gujarat Region Studies

Limited studies specifically from Gujarat region emphasize the need for regional data:

Patel et al. (2014) (Gujarat):

- DNS prevalence: 70%
- Concha bullosa: 28%
- Showed variations from national averages

Importance of Regional Studies:

Regional variations exist due to genetic, environmental, and anthropometric differences. South Gujarat has a unique demographic mix requiring dedicated study.

8. CLINICAL SIGNIFICANCE OF ANATOMICAL VARIATIONS

8.1 Surgical Implications

Stammberger (1991) and Kennedy (1992) emphasized:

- Anatomical variations predispose to chronic rhinosinusitis
- Pre-operative CT mapping essential for FESS
- Recognition of variations prevents surgical complications

8.2 Complications Related to Anatomical Variations

Anatomical Variation | Potential Complication

Dehiscent lamina papyracea | Orbital injury

Low Keros Type III | CSF leak, meningitis

Onodi cell | Optic nerve injury

ICA dehiscence | Carotid artery injury

Haller cells | Maxillary ostium obstruction

Concha bullosa | OMC obstruction

May et al. (1994) reported complication rates of 0.3-3% in FESS, many related to unrecognized anatomical variations.

8.3 Association with Sinus Disease

Bolger et al. (1991) correlated variations with sinusitis:

- Concha bullosa: Associated with ipsilateral maxillary sinusitis
- Haller cells: Associated with recurrent maxillary sinusitis
- DNS: Associated with contralateral sinus disease

Caughey et al. (2005) found statistically significant association between anatomical variations and chronic rhinosinusitis.

9. IMAGING MODALITIES COMPARISON

Modality | Advantages | Disadvantages

Plain X-ray | Low cost, available | Poor detail, superimposition

CT | Gold standard, bone detail, mu | Radiation exposure

MRI | Soft tissue detail, no radiati | Poor bone detail, cost

Cone Beam CT | Low dose, good bone detail | Limited soft tissue contrast

10. SUMMARY TABLE: PREVALENCE OF ANATOMICAL VARIATIONS

Anatomical Variation | Western Studies | Indian Studies

Deviated Nasal Septum | 20-79% | 58-75%

Concha Bullosa | 24-53% | 25-32%

Paradoxical Middle Turbinate | 17-27% | 12-15%

Agger Nasi Cells | 89-98% | 87-92%

Haller Cells | 10-45% | 15-18%

Onodi Cells | 8-14% | 10-12%

Frontal Sinus Aplasia | 5-15% | 3.8-4.2%

Keros Type III | 8-26% | 15-20%

ICA Dehiscence | 4-22% | 7-8%

Accessory Maxillary Ostium | 10-30% | 18%

11. GAPS IN LITERATURE

1. Regional Data Deficiency: Limited CT-based studies from South Gujarat region
2. Population-Specific Variations: Need for anthropometric correlation
3. Bilateral Asymmetry: Insufficient documentation of side-to-side variations
4. Age and Gender Correlation: Need for detailed demographic analysis
5. Clinical Correlation: Association with chronic rhinosinusitis patterns in regional population

12. RATIONALE FOR PRESENT STUDY

The review of literature reveals:

1. Significant variations exist in paranasal sinus anatomy across different populations
2. Regional differences within India necessitate localized studies
3. CT scan remains the gold standard for anatomical evaluation
4. Pre-operative knowledge of variations is crucial for safe FESS
5. South Gujarat region lacks comprehensive CT-based anatomical data

Therefore, the present study aims to:

- Document anatomical variations of paranasal sinuses in South Gujarat population

- Establish regional baseline data for surgical planning
- Correlate variations with demographic parameters
- Compare findings with existing Indian and international literature

13. KEY REFERENCES

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This review of literature provides the foundation for understanding anatomical variations of paranasal sinuses and establishes the need for a CT-based study in the South Gujarat population.

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Topic: Anatomical Variation of Paranasal Air Sinuses: A CT Study in South Gujarat Region